

Head MR Angiography (MRA)

ACG: A-0033 (AC)

[Link to Codes](#)

This MCG care guideline may overlap CMS guidance located in CMS statutes and regulations that are not currently part of MCG's Medicare Compliance content. For Medicare Advantage beneficiaries, before referring to MCG care guidelines, users should first follow all guidance related to this topic located in CMS National Coverage Determinations (NCDs), Local Coverage Determinations (LCDs), and other statutes and regulations. Content used in MCG care guidelines may be used to supplement but does not replace, modify, or supersede existing Medicare regulations or applicable NCDs or LCDs.

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Clinical Indications

- Head MR angiography (MRA) may be indicated for **1 or more** of the following(1)(2)(3)(4)(5):
 - ☐ Aneurysm or arteriovenous malformation, known or suspected, as indicated by **1 or more** of the following(23)(24)(25)(26)(27):
 - ☐ Follow-up monitoring for patient with history of subarachnoid hemorrhage(33)(37)
 - ☐ Follow-up of known aneurysm or arteriovenous malformation(38)(39)
 - ☐ Initial evaluation of suspected aneurysm or arteriovenous malformation (eg, acute severe headache or diagnosis of intracranial or subarachnoid hemorrhage)(39)(40)(41)(42)(43)
 - ☐ Intracerebral hemorrhage, known or suspected(44)(45)
 - ☐ Re-evaluation after surgical or endovascular therapy (including coiling) for cerebral artery aneurysm or arteriovenous malformation(33)(36)(37)(46)
 - ☐ Intracranial arterial dissection, suspected(47)(48)(49)☐
 - ☐ Intracranial arterial occlusive disease, suspected, when imaging results will impact patient management(51)(52)(53)(54)(55)(56)☐
 - ☐ Intracranial hypertension (idiopathic), suspected(65)(66)☐
 - ☐ Intracranial vasculitis, suspected(27)(68)(69)(70)(71)(72)☐
 - ☐ Intracranial venous occlusive disease, suspected(78)(79)(80)(81)(82)☐
 - ☐ Preoperative or preprocedural planning for carotid endarterectomy or percutaneous intervention needed, as indicated by **1 or more** of the following(92)(93)(94):☐
 - ☐ Acute stroke, and performed in conjunction with neck MRA
 - ☐ Duplex (Doppler) scan finding of high-grade stenosis in asymptomatic patient
 - ☐ Duplex (Doppler) scan findings indeterminate
 - ☐ Duplex (Doppler) scan or clinical findings show high carotid bifurcation, loops, or kinks.
 - ☐ History of neck radiation therapy
 - ☐ Preoperative planning for delineation of vascular supply of vascular neoplasm(3)(95)☐
 - ☐ Screening for cerebral artery aneurysm in patient with **1 or more** of the following(25)(46)(60):☐
 - ☐ Autosomal dominant polycystic kidney disease(34)(97)(98)
 - ☐ Two or more family members with intracranial aneurysm or subarachnoid hemorrhage
 - ☐ Tinnitus, pulsatile(99)(100)☐
 - ☐ Traumatic injury to cervicocerebral vessels, suspected(88)(101)(102)(103)(104)☐
 - ☐ Vascular malformation of brain or skull base, for confirmation of diagnosis or preoperative planning(52)(106)(107)(108)(109)☐

- ☐ Repeat evaluation of specific area or structure with same imaging modality, and **ALL** of the following:
 - Clinical need for repeat imaging, as indicated by **1 or more** of the following:
 - Change in clinical status (eg, worsening symptoms or new associated symptoms), and findings may impact treatment
 - Need for interval reassessment that may impact treatment plan
 - Need for re-imaging either prior to or after performance of invasive procedure
 - Prior imaging results of specific area or structure with same imaging modality documented and available for comparison

Evidence Summary

Background

MRA uses MRI technology to produce detailed 2-dimensional or 3-dimensional images of the vascular system and perivascular anatomy without the risks usually associated with catheter angiography, including radiation exposure and arterial puncture.(6) **(EG 2)** The gadolinium-based contrast material used for MRA carries a lower risk of allergic reaction than the iodine-based contrast material used in catheter angiography and CTA.(7) **(EG 2)** Some gadolinium-based contrast agents used for diagnostic and treatment guidance with MRI or MRA have been associated with nephrogenic systemic fibrosis in patients with chronic kidney disease or acute kidney injury.(7)(8)(9)(10)(11) **(EG 2)** Gadolinium deposition in the brain has been reported by multiple research groups; however, the clinical significance of this finding is unclear.(10)(11)(12)(13) **(EG 2)** Specialty society guidelines recommend weighing the risks vs benefits when using gadolinium-based contrast agents for MRI.(7)(14) **(EG 2)** Time-of-flight MRA is a technology that exploits the difference in MRI signal intensities between flowing vs static blood, thus creating an image of blood vessels without using intravenous contrast.(5) (15)(16) **(EG 2)** MRI-based techniques may be contraindicated in patients with ferromagnetic intracranial aneurysm clips and certain types of ferromagnetic foreign bodies and cochlear implants.(17) **(EG 2)** Although these implants have traditionally included cardiac pacemakers and implantable cardioverter-defibrillators, multiple trials have validated the safe and effective use of pacemaker systems designed for MRI use, and a guideline supported by multiple specialty societies states that many patients with implantable cardiac devices, including those not originally intended for MRI scanning, can safely undergo MRI scanning under defined clinical indications. (18)(19)(20)(21)(22) **(EG 1)**

Criteria

The evidence for the clinical indications found in this guideline includes 86 published peer reviewed articles, 13 specialty society or other evidence-based guidelines, 1 Cochrane systematic review, and 1 book section.

For cerebral artery aneurysm or arteriovenous malformation, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Catheter angiography remains the gold standard for diagnosis and formal delineation of cerebral artery aneurysms or arteriovenous malformations; however, MRA is being used more frequently as a noninvasive alternative to catheter angiography, particularly as magnetic resonance technology has improved.(28) **(EG 2)** For diagnosing cerebral aneurysms, a meta-analysis of 10 studies comparing CTA with MRA found that CTA had a pooled sensitivity of 84% and specificity of 85% and MRA had a pooled sensitivity of 80% and specificity of 87%; the authors concluded that both CTA and MRA have high accuracy for diagnosing cerebral aneurysms.(29) **(EG 1)** A meta-analysis of 18 studies with 3463 patients with or without subarachnoid hemorrhage found that 3-dimensional time-of-flight MRA had a sensitivity of 89% and a specificity of 94% when compared with digital subtraction angiography for the detection of intracranial aneurysms.(30) **(EG 1)** A systematic review and meta-analysis of 12 studies encompassing 960 patients with intracranial aneurysms found that pooled sensitivity of MRA was 95% and pooled specificity was 89% for the detection of intracranial aneurysms.(31) **(EG 1)** A prospective study of 23 patients with arteriovenous malformations found complete agreement between MRI and conventional catheter angiography in evaluating patency of the arteriovenous malformations.(32) **(EG 2)** Furthermore, MRA is a first-line option for screening patients with a history of intracranial aneurysms (especially with a history of rupture) or autosomal dominant polycystic kidney disease, or who have a first-degree relative with a history of subarachnoid hemorrhage or intracranial aneurysms.(23)(33)(34) **(EG 2)** A review article notes that MRA is a reliable tool for confirming secondary causes of intracerebral hemorrhage. This review article also notes that MRA is useful for distinguishing between primary intracerebral hemorrhage and hemorrhagic transformation of acute ischemic stroke.(35) **(EG 2)** A meta-analysis of 43 studies found that time-of-flight MRA and contrast-enhanced MRA had sensitivity of 88% and 88%, respectively, and specificity of 94% and 96%, respectively, for the detection of recanalization after treatment of coiled or flow-diverted intracranial aneurysms when compared with digital subtraction angiography as the reference standard.(36) **(EG 1)** A specialty society guideline recommends head MRA with contrast after aneurysm coil or stent placement; metal in the coil or stent may distort noncontrast CTA and obscure residual abnormality.(33) **(EG 2)**

For intracranial arterial dissection, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review noted that initial evaluation of carotid and vertebral artery dissection is usually made by MRI in combination with MRA or CTA; the authors concluded that test characteristics among the 3 imaging modalities were relatively similar for the diagnosis of carotid and vertebral artery dissection.(50) **(EG 2)** A specialty society guideline and appropriate-use criteria note that head MRA is appropriate for the evaluation of cervicocerebral vascular dissection.(3)(40) **(EG 2)**

For intracranial arterial occlusive disease, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** MRA and CTA are appropriate initial imaging techniques for detection of significant stenoses, and reviews have reported sensitivity and specificity of close to 100%.(57) **(EG 2)** Review articles note that CTA and MRA of the head and neck are useful imaging modalities for diagnosis and management of acute ischemic stroke in children and adults.(16)(58)(59)(60)(61) **(EG 2)** A single-center retrospective

study of 82 neonates with arterial ischemic stroke identified abnormal brain MRA findings in 29 (35%) neonates; risk factors for abnormal arterial findings were low Apgar scores and sepsis. The authors recommended brain MRA for neonates with ischemic strokes, especially those with sepsis or low Apgar scores.(62) **(EG 2)** Practice guidelines recommend that MRA or CTA be available at primary stroke centers for admitted stroke patients who may benefit from such testing when needed to guide or determine treatment decisions; CTA or MRA is recommended as part of the initial evaluation of stroke patients who may be candidates for endovascular intervention.(33)(54)(63)(64) **(EG 2)**

For intracranial hypertension (idiopathic), evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Review articles note that MRA with venography is used to exclude venous sinus thrombosis as a secondary cause of elevated intracranial pressure.(66)(67) **(EG 2)**

For intracranial vasculitis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Review articles note that MRA, due to its ability to produce detailed images of the vasculature, can be a useful diagnostic test for cerebral vasculitis as well as other central nervous system vasculitides in both children and adults.(73)(74)(75)(76)(77) **(EG 2)** Appropriate-use criteria state that MRA is appropriate in children with known or suspected vasculitis and symptoms of acute stroke.(40) **(EG 2)**

For intracranial venous disease (eg, venous sinus thrombosis), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A specialty society guideline states that CTA and MRA are appropriate first-line imaging options for evaluation of intracranial venous occlusive disease; venous MRA is an appropriate imaging option for intracranial venous disease.(79) **(EG 2)** However, catheter angiography remains the definitive diagnostic reference standard and is appropriate whenever results from noninvasive imaging are indeterminate, or when endovascular intervention or thrombolysis is being considered.(78)(83)(84)(85)(86) **(EG 2)** A review article states that MRA is superior to CTA for detecting thrombosis in the cortical or deep cerebral veins.(87) **(EG 2)** A review article indicates that venous MRA may be used to identify posttraumatic venous sinus occlusion, sinus laceration, or cortical vein injury when iodinated contrast is contraindicated.(88) **(EG 2)** Venous MRA as a dedicated neuroimaging technique can be crucial for diagnosis of dural sinus venous thrombosis in neonates because the clinical presentation is nonspecific.(89)(90) **(EG 2)** Review articles note that MR venography can be useful for detection and characterization of cerebral sinovenous thrombosis in pediatric patients.(89) (91) **(EG 2)** Appropriate-use criteria state that MRA is appropriate in children with suspected cortical vein or dural sinus thrombosis.(40) **(EG 2)**

For preprocedural planning for carotid endarterectomy or percutaneous intervention, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline notes that, when intervention is planned for significant carotid stenosis detected by duplex sonography, CTA, MRA, or digital subtraction angiography of the head and neck can be useful to evaluate stenosis severity and to identify intracranial or intrathoracic vascular lesions that are inadequately assessed by duplex sonography.(92) **(EG 2)**

For preoperative planning for delineation of vascular supply of vascular neoplasm, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Specialty society guidelines state that head MRA is appropriate in adult and pediatric patients for evaluation of vascular supply to tumors, particularly for the purposes of preoperative planning.(3)(95) **(EG 2)**

For screening for cerebral aneurysms, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A meta-analysis of 15 studies with 1490 patients with autosomal dominant polycystic kidney disease who had brain imaging with either MRA, CTA, or digital subtraction angiography found that 10% had intracranial aneurysms; a family history of intracranial aneurysm or hemorrhagic stroke significantly increased the risk of aneurysm occurrence.(96) **(EG 1)** A prospective cohort trial of 495 patients with autosomal dominant polycystic kidney disease who were offered head MRA screening found that 14% of those with risk factors (ie, personal or familial history of intracranial aneurysm) had an unruptured aneurysm as compared with 6% of those without risk factors. However, at a median of 5.9 years, the rate of intracranial aneurysm rupture was the same for patients with and without risk factors (2 per 1000 patient-years); the authors concluded that screening for intracranial aneurysms should be considered for all patients with autosomal dominant polycystic kidney disease.(97) **(EG 2)** A specialty society guideline recommends brain imaging with either MRA or CTA to screen for unruptured aneurysms in patients with either a history of autosomal dominant polycystic kidney disease (especially those with a family history of intracranial aneurysms) or 2 or more family members with intracranial aneurysms or subarachnoid hemorrhages.(25) **(EG 2)**

For pulsatile tinnitus, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** For patients with symptoms of pulsatile tinnitus, specialty society guidelines support imaging, including MRA of the head, to evaluate for possible carotid vascular lesions (eg, atherosclerosis, dissection, fibromuscular dysplasia) or intracranial vascular etiologies (eg, arteriovenous malformation, dural arteriovenous fistula, vascular tumor).(99)(100) **(EG 2)**

For traumatic injury to cervicocerebral vessels, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Specialty society guidelines state that MRA is appropriate for the evaluation of cervicocerebral vascular trauma, including dissections, occlusions, or pseudoaneurysms.(3) (105) **(EG 2)**

For vascular malformations of the brain or skull base, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review of 3 studies that compared MRA with catheter-based digital subtraction angiography for the detection of intracranial vascular malformations following intracerebral hemorrhage found that MRA had a pooled sensitivity of 98% and specificity of 99%; however, the authors noted that the included studies had methodological shortcomings, such as retrospective design and verification bias, that may have increased apparent test accuracy.(110) **(EG 1)** A review article notes that MRA is the preferred imaging modality for the diagnosis of pediatric moyamoya.(109) **(EG 2)** A specialty society guideline and appropriate-use criteria state that MRA is appropriate for the evaluation of congenital or acquired vascular abnormalities.(3)(40) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(111); 42 CFR 422.101(112)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(113); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(114); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(115)

Medicare Coverage Determinations: Medicare Coverage Database(116)

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